Attention Is All You Need

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Abstract: The dominant sequence transduction models are based on RNNs.

We propose the Transformer, based solely on attention mechanisms.

1 Introduction

Recurrent models preclude parallelization within training examples.

3 Model Architecture

The encoder is composed of a stack of N = 6 identical layers.

Each layer has multi-head self-attention and position-wise feed-forward.

In this work we employ h = 8 parallel attention layers, or heads.

For each of these we use d_k = d_v = d_model/h = 64.

5 Training and Results

We trained on the standard WMT 2014 English-German dataset consisting of 4.5M sentence pairs.

We used the Adam optimizer with beta_1 = 0.9, beta_2 = 0.98 and epsilon = 10^-9.

We used warmup_steps = 4000 with inverse square root decay.

We trained base models on one machine with 8 NVIDIA P100 GPUs.

On WMT 2014 En-De, the Transformer achieves 28.4 BLEU.